

Gastrointestinal Disorders and Skin Manifestations

Gardner Syndrome

Gardner syndrome is an autosomal dominant disorder characterized by cutaneous cysts and pre-malignant adenomatous polyps, particularly in the colorectum. Duodenal and hepatic carcinomas can also occur. The cutaneous lesions frequently precede gastrointestinal involvement and include solitary or multiple epidermoid cysts, fibromas, lipomas, pilomatricomas, facial osteomas, and distinctive ocular lesions.

Peutz-Jeghers Syndrome

Peutz-Jeghers syndrome is an autosomal dominant condition marked by small, darkly pigmented macules around the mouth, lips, buccal mucosa, and digits. It is associated with a unique type of hamartomatous polyp, primarily in the small intestine, which predisposes to recurrent intussusception. Gastrointestinal bleeding is relatively uncommon, but up to 2% of patients develop adenocarcinoma of the stomach, duodenum, or colon. There is also a generalized increased risk of malignancy.

Cronkhite-Canada Syndrome

Cronkhite-Canada syndrome presents with patchy alopecia and characteristic nail changes. Patients have inflammatory polyps in the stomach and bowel, along with abdominal pain and severe protein-losing enteropathy.

Inflammatory Bowel Disease (Ulcerative Colitis and Crohn Disease)

Both ulcerative colitis and Crohn disease manifest with abdominal pain, gastrointestinal bleeding, or diarrhea. Ulcerative colitis predisposes to colorectal carcinoma.

Skin Complications

The cutaneous manifestations of both conditions are similar, though they differ in frequency: - **Pyoderma gangrenosum** occurs in up to 50% of patients with this condition, many of whom have inflammatory bowel disease. However, only 12% of patients with ulcerative colitis and 2% with Crohn disease develop pyoderma gangrenosum. The severity of ulceration often correlates with disease activity and may resolve with effective treatment of the underlying bowel disease. - **Erythema nodosum** is a well-recognized extraintestinal manifestation.

Specific Features in Crohn Disease

- **Oral granulomatous nodules** are common and may coalesce to form a "cobblestone" appearance.
- **Granulomatous cheilitis** may precede intestinal symptoms by several years.
- Granulomas can appear in the perineum, at stoma sites (colostomy or ileostomy), and within scars, sinuses, and fistulas.
- Rarely, granulomas develop in non-contiguous areas such as the trunk and limbs—termed "metastatic" cutaneous Crohn disease.
- **Aphthous ulcers** may be an early presenting feature.

Pancreatic Disease

- **Cullen's sign** (periumbilical bruising) and **Grey Turner's sign** (flank bruising) are recognized signs of acute pancreatitis, though they occur in less than 3% of cases.
- **Chronic pancreatitis** can lead to tender subcutaneous nodules due to fat necrosis, which may ulcerate.

Gastric Bypass Surgery

Gastric bypass surgery, including procedures with jejunoileal or jejunocolic anastomoses, is used to treat morbid obesity. Postoperative complications include: - Dry skin - Hair loss - Recurrent fever - Arthralgia - Inflammatory and vasculitic skin lesions

A similar clinical picture, known as "bowel bypass syndrome without bowel bypass," can occur in the context of other gastrointestinal disorders.

Cutaneous Effects of Malabsorption of Specific Nutrients

Vitamin deficiencies may result from intestinal surgery, pancreatic disease, malabsorption syndromes, or malnutrition (including alcohol-related malnutrition).

Vitamin A Deficiency

Causes xerotic, wrinkled skin covered with fine scales, sometimes accompanied by deep erosions (dermomalacia).

Vitamin B₃ (Niacin) Deficiency

Leads to **pellagra**, characterized by the triad of: - Diarrhea - Dermatitis - Dementia

Skin Changes Due to Malabsorption

Non-specific manifestations:

- Acquired ichthyosis
- Pruritus
- Hair and nail changes
- Hyperpigmentation
- Altered skin texture and elasticity
- Eczematous or psoriatic rashes
- Associated with jejunoileal and jejunocolic anastomoses

Specific nutrient deficiencies:

- Zinc
- Essential fatty acids
- Vitamins

Malabsorption Due to Skin Disease

- Dermatogenic enteropathy
- Collagen vascular disease
- Dermatitis herpetiformis and celiac disease